SYLLABUS

1. Course Information

Course Code:	CCC101
Course Title:	Computer Programming 1
Description:	The course covers the use of general-purpose programming language to solve problems. The emphasis is to train students to design, implement, test, and debug programs intended to solve computing problems using fundamental programming constructs.
Credits:	3 units
Modality:	Face-to-face
Contact Hours:	18 weeks (January 12, 2026 to , May 11, 2025)
Pre-requisite / Co-requisite:	None
A.Y. and Semester:	2nd Semester, 2025-2026

2. Course Outcomes

At the end of the semester, the students should be able to:

CO1: Design, implement, test, and debug program based on a given specification that uses each of the following fundamental components: (1) primitive data types, (2) basic computation, (3) simple I/O, (4) conditional and iterative structures, (5) definition of functions and parameter passing, and (6) recursion
CO2: Analyze and explain the behavior of simple programs involving the fundamental programming contructs, variables, expressions, assignments, I/O, control constructs, functions, parameter passing, and recursion.

3. Alignment of Course Outcomes (COs) to Program Outcomes (POs)

The COs are expected to support the achievement of the following program outcomes (POs).

a) Analyze complex problems, and identify and define the computing requirements needed to design an appropriate solution.
b) Apply computing and other knowledge domains to address real-world problems.
c) Ability to apply knowledge of computing, basic science, and mathematics appropriate to the discipline and the program educational objectives.
d) Ability to analyze a problem, identify and define the computing requirements appropriate to the problem's solution.

4. Learning Plan

Week	Topic	Intended Learning Outcome	Teaching and Learning Activities	Learning Resources	Assessment Strategies/Tools	Evidence of Outcome
1	Introduction: Programming Basics	At the end of the unit, the student is able to write C program using basic input and output.	Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	A simple computer program with input and output statements.

2	Functions - Definitions and Prototypes	At the end of the unit, the student is able to create Basic C program using functions and prototypes.	Lecture	PowerPoint presentation Demo program	Quiz	An application using the fundamental types as well as arithmetic operators applying preced- ence rules.
3	Functions - Scope Rules	At the end of the unit, the student is able to create Basic C program using functions and prototypes.	Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Application solving a given problem using simple decision making statements.
4	Exam 1					

Table 1

Week	Topic	Intended Learning Outcome	Teaching and Learning Activities	Learning Resources	Assessment Strategies/Tools	Evidence of Outcome

5	Decision Control Structures: if, if-else, switch	Understand different types of decision control structures and apply to certain problem.	Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Application solving a given problem using repetition statements.
6	Repetition Control Structures: while, dowhile, for	Understand different types of repetition control structures and apply to certain problem.	Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Application on the use of increment, decrement, and assignment operators.
7	Arrays	create representation of certain problems involving lists and string input.	Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Program using array data structure to represent a set of related data items.
8	Arrays - Linear Search	Use arrays in solving searching problem.	Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Demonstration on how to pass arrays to functions.

Week	Topic	Intended Learning Outcome	Teaching and Learning Activities	Learning Resources	Assessment Strategies/Tools	Evidence of Outcome
9	Arrays – Sorting	Use arrays in solving sorting problem.	Lecture, Program Demo, Guided Programming		Quiz, Programming Exercise	Basic searching application.
10	Exam 2					
11	Recursion	Apply recursion to certain mathematical problems.	Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Basic sorting application.
12	Pointers		Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Use pointers to pass arguments to functions by reference.

13	Strings		Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Program that use each of the following data structures: arrays, records, and strings.
14	File Processing		Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Program that use each of the following data structures: arrays, records, and strings.
15	Structures		Lecture, Program Demo, Guided Programming	PowerPoint presentation Demo program	Quiz, Programming Exercise	Program that use each of the following data structures: arrays, records, and strings.
16	Exam 3					

5. Grading System

Table 2

Criteria	**Weight**
Major Exams	30%
Quizzes/Seatwork/Class Participation	30%
Project	35%
Attendance	5%
Total	100%

Passing Rate = 50%

6. References

Hanly, Jeri B. & Koffman, Elliot B.,(2005) Problem Solving and Program Design in C. 5th Ed. Temple University, Pearson Education, Inc., Boston, MA, USA

7. Classroom Policies

Please refer to Students' Handbook.

8. Approval

Prepared by:	**Approved:**	
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Faculty Member	Department Chairperson	Dean